Seminar Talk 4;
Blow-up, genus and some sheaves of differentials

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Abstract

This talk is divided into two parts. The first will be dedicated to the notion of genus, while the second will introduce the procedure of "blowing-up" in scheme-theory. The genus will come in two flavors - The arithmetic and the geometric genus - and they will coincide in nice scenarios. To introduce them we will revisit a few sheaves related to differential calculus. The second topic, being the Blow-up of a scheme at a closed subscheme will formalizes the idea that there is a universal way to make the defining equations of that subscheme locally principal. Finally, we will have time for some exercises to wrap up.

1 Genus (and a quick review of differentials)

Definition 1.1. *recall:* Module of relative differential forms (affine case):
The A-module $\Omega_{A/k}$ is also called the module of *Käler differentials* on A (relative to k; also possible for general B-algebras) and is generated in the following way:

1. Generators: Symbols $d(a)$ and a with $a \in A$.
2. Relations: $d(ta) = td(a)$, $d(a_1a_2) = a_1d(a_2) + a_2d(a_1)$ and $d(a_1 + a_2) = d(a_1) + d(a_2)$ for $t \in k$ and $a, a_1, a_2 \in A$.

Remark. In less explicit terms this module is determined via a universal property for certain maps $D : A \to M$ (satisfying identities encoding the above relations: additivity, annihilates the image of k in A, the Leibniz rule $D(ab) = aD(b) + bD(a)$). Hence it comes with an *k-linear derivation* $d : A \to \Omega_{A/k}$; $a \mapsto d(a)$. Every such derivation $D : A \to M$ factors uniquely through d:

$$\mathrm{Hom}_A(\Omega_{A/k}, M) \xrightarrow{\simeq} \mathrm{Der}_k(A, M);\ \alpha \mapsto \alpha \circ d$$

(k-linearity is equivalent to annihilates the image of k in A assuming Leibnitz:
$d(1) = d(1 \cdot 1) = 1 \cdot d(1) + 1 \cdot d(1) = 2d(1)$ hence $d(1) = 0$ and thus $d(t) = d(1 \cdot t) = t \cdot d(1) = 0$.)

Lemma 1.2. *"compatibility on overlaps"*
There is a canonical isomorphism: $\Omega_{A/k} \otimes_A A_f \simeq \Omega_{A_f/k}$.

Proof. Given a module M over the localization A_f, we have a natural isomrophism between derivations $\mathrm{Der}_k(A, M) \simeq \mathrm{Der}_k(A_f, M)$, since we can extend a derivation via the quotient rule to the localization (i.e. setting $a/s \mapsto (sd(a) - ad(s))/s^2$) and restrict it along the localization map back. Hence we get a chain of isomorphisms yielding us the desired isomorphism via Yoneda:

$$\mathrm{Hom}_{A_f}(\Omega_{A/k} \otimes_A A_f, M) \simeq \mathrm{Hom}_A(\Omega_{A/k}, M) \simeq \mathrm{Der}_k(A, M) \simeq \mathrm{Der}_k(A_f, M) \simeq \mathrm{Hom}_{A_f}(\Omega_{A_f/k}, M).$$

□

Definition 1.3. Sheaf of relative differential forms (general case):
For a scheme $X \to \operatorname{Spec} k$, we define the quasi-coherent sheaf $\Omega_{X/k}$ by descent from the collection of quasi-coherent sheaves $\{\Omega_{A_i/k}\}_i$ on an affine open cover $\{\operatorname{Spec} A_i\}_i$ together with the above isomorphisms on the overlaps 1.2.

Corollary 1.4. *For $X \to \operatorname{Spec} k$ smooth of dimenston r, the sheaf $\Omega_{X/k}$ is locally free of rank r.*

Proof. This follows from a more general relative version (*cf.*: [LE02], Section 6.2.1. Prop 2.5). (In addition there is a nice to know stalk-wise smoothness criteria (*cf.*: op. cit. Prop 2.2).) □

Definition 1.5. The canonical bundle:
For a smooth scheme $X \to \operatorname{Spec} k$ of dimension r we define the *canonical bundle* (relative to k) to be the exterior power $\omega_{X/k} := \wedge^r \Omega_{X/k}$.

Remark. Since for a smooth scheme $\Omega_{X/k}$ is a rank r vector bundle on X, we get that $\omega_{X/k}$ is a line bundle. Moreover if X is a curve, we just recover $\Omega_{X/k}$.

Remark. Often the notation $\Omega^r_{X/k} := \wedge^r \Omega_{X/k}$ is used for a general r, called the *sheaf of differentials of order r* (and hence one would also use the notation $\Omega^1_{X/k}$ instead of simply $\Omega_{X/k}$).
There will therefore be a more general notion of the *canonical bundle* $\omega_{X/k}$ ((*cf.*: [LE02], Section 6.4.2 Definition 4.7.) that in our smooth setting will coincide with our choice of definition being the differentials of "top order".

Lemma 1.6. *Duality:*
For a smooth, propper $f : X \to \operatorname{Spec} k$ of dimension r, we have $\mathrm{H}^0(X, \omega_X) \simeq \mathrm{H}^r(X, \mathcal{O}_X)^\vee$.

Proof. This follows from the more general *Grothendieck duality* which we will not proof (*cf.*: [LE02] 6.4.3; and for a special case the talk about Serre duality in the last seminar "Cohomology of coherent shheaves"):

$$f_* \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F}, \omega_f^\circ) \simeq \mathcal{H}\mathrm{om}_{\mathcal{O}_Y}(\mathbf{R}^r f_* \mathcal{F}, \mathcal{O}_Y)$$

for a propper morphism $f : X \to Y$ to a locally Noetherian scheme Y with fibers of dimension $\leq r$ which is the dimension of X in our case. Also ω_f° (the *dualizing sheaf*) coincides in our smooth setting with ω_X.
In our case with $\mathcal{F} = \mathcal{O}_X$ we get by definition that $f_* \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{O}_X, \omega_X) =: \mathrm{H}^0(X, \omega_X)$ and $\mathcal{H}\mathrm{om}_{\mathcal{O}_Y}(\mathbf{R}^r f_* \mathcal{O}_X, \mathcal{O}_Y) = \mathrm{Hom}_k(\mathrm{H}^r(X, \mathcal{O}_X), k) = \mathrm{H}^r(X, \mathcal{O}_X)^\vee$ □

Lemma 1.7. *Finiteness of cohomology:*
For a proper morphism $f : X \to \operatorname{Spec} A$, A notherian (for us k) and a coherent sheaf $\mathcal{F}$ on X we have that all $\mathrm{H}^i(X, \mathcal{F})$ are finite A-modules.

Proof. (*cf.*: [Sta23], Tag 02O6, Lemma 30.19.2.) □

Lemma 1.8.
For $X \to \operatorname{Spec} k$ proper, geometrically integral, we get $\mathrm{H}^0(X, \mathcal{O}_X) \simeq k$.

Proof. (*cf.*: [Sta23], Tag 0BUG, Lemma 33.9.3. (8)) By 1.7 we know that $\mathrm{H}^0 := \mathrm{H}^0(X, \mathcal{O}_X)$ is a finite k-vector-space. We also know that $\mathrm{H}^0 = \mathcal{O}_X(X)$ is geometricsally integral (i.e. $\mathrm{H}^0 \otimes_k \overline{k}$ is integral; this is not trivial, use flat basechange) and in particular integral. Since that means that the multiplication on H^0 is injective (and by Rank-Nullity also surjective, so every element is invertible), we get H^0 a field (extension of k).
But H^0 then has to be k since $\mathrm{H}^0 \otimes_k \overline{k}$ (which is integral and hence by a similar argument a finite field extension of $\overline{k}$) is just $\overline{k}$ and hence $[\mathrm{H}^0 : k] = \dim_{\overline{k}} \mathrm{H}^0 \otimes_k \overline{k} = 1$. □

Definition 1.9. Geometric Genus:
The *geometric genus* of the smooth, proper, geometrically integral curve $C \to \operatorname{Spec} k$ is defined to be the integer

$$g(C) := p_g(C) := \dim_k \mathrm{H}^0(C, \omega_{C/k}).$$

In this smooth setting (by our definition of $\omega_{C/k} := \Omega^n_{C/k}$)) and the fact that C is a curve ($\dim = 1$) we have that this is the same as $\dim_k \mathrm{H}^0(C, \Omega^1_{C/k})$.
By the duality 1.6 and 1.7 we also get: $\dim_k \mathrm{H}^0(C, \omega_{C/k}) = \dim_k \mathrm{H}^1(C, \mathcal{O}_C)^\vee = \dim_k \mathrm{H}^1(C, \mathcal{O}_C)$.

Remark. $\dim_k \mathrm{H}^1(C, \mathcal{O}_C)$ is obviously defined for more general C.
This definition though nicely resembling the the topology of the scheme under the above assumptions (e.g. over $\mathbb{C}$ it connects to the genus of the Riemann surfact $C(\mathbb{C})$) does not admit the nicest formal properties that one might hope for.
To illustrate this we can look at a proper flat family $C \to B$, such that some smooth special fiber $C_{t\neq 0}$ "degenerates" into a singular special fiber $C_{t=0}$. One can observe that here the dimension of H^1 might jump. An example is given in ([Har77], chapter III, Example 12.9.2. referring to:)

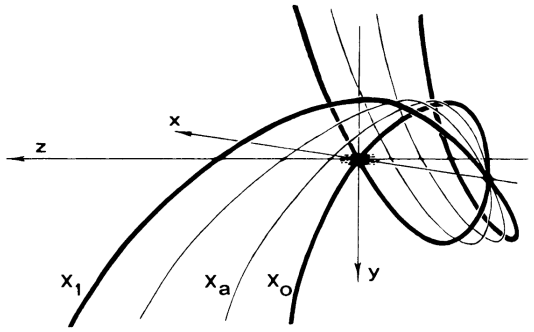


Figure 11. A flat family of subschemes of $\mathbf{P}^3$.

Figure 1: [Har77], chapter III, Example 9.8.4.
(An *embedded point* in the singular fiber makes the dimension of H^0 jump)

The reason in a sense is, that there is a hidden term, namely $1 - \dim_k \mathrm{H}^0(C, \mathcal{O}_C)$ that vanishes in our setting 1.8 that we could add to our definition (as exactly the error-term that will make the resulting notion constant in families, for this (*cf.*: [Har77] chapter III, Thm 9.9) which is slightly more general via Hilbert polynomials) motivating the following definition of *arithmetic genus*.

Definition 1.10. (*Recall*) Euler-Poincaré characteristic:
Given a coherent sheaf $\mathcal{F}$ on C, we define the *Euler-Poincaré characteristic* to be

$$\chi_k(\mathcal{F}) := \chi_k(C, \mathcal{F}) := \sum_{i \geq 0} (-1)^i \dim_k \mathrm{H}^i(C, \mathcal{F}).$$

Remark. This definition will obviously still be reasonable for more general schemes as long as finiteness is guaranteed as in 1.7 (the vanishing for all $i > n$ for a topological space of dim n is also well known; *cf.: Grothendiecks vanishing:* [Sta23], Tag 02UU, Prop. 20.20.7.).

Definition 1.11. Arithmetic Genus:
The *arithmetic genus* of the proper curve C is defined to be the integer

$$p_a(C) := p_a(C/k) := 1 - \chi_k(\mathcal{O}_C)$$

which enjoins the fact that when ever $\dim_k \mathrm{H}^0(C, \mathcal{O}_C) = 1$ and in particular in the above setting it reduces to $p_a(C) = \dim_k \mathrm{H}^1(C, \mathcal{O}_C) = p_g(C)$.

Remark. The definition of p_a is dependent on k. But for an extension k'/k and for the basechange $C_{k'}$ of C to k' we have that $p_a(C_{k'}/k') = p_a(C/k)$ (this essentially follows from *flat basechange* since field extensions are flat, for example see [LE02] Corollary 5.2.27; tensoring with a flat module "preserves the exactness" of the Čech complex).

Example. For $C = \mathbb{P}^1_k$ we know that $g = p_g(C) = p_a(C)$ (p_g is a special case of p_a as the conditions apply). Moreover $\mathrm{H}^1(C, \mathcal{O}_C)$ vanishes, hence $g = 0$.

To see this we can use the Čech complex that computes cohomology. Recall this is given by:

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{O}_{\mathbb{P}^1_k}) := [0 \to \check{\mathcal{C}}^0 := \mathcal{O}_{\mathbb{P}^1_k}(U_0) \times \mathcal{O}_C(U_1) = k[x] \times k[x^{-1}] \xrightarrow{d} \check{\mathcal{C}}^1 := \mathcal{O}_{\mathbb{P}^1_k}(U_0 \cap U_1) = k[x, x^{-1}] \to 0]$$

where the cover $\mathcal{U} = \{U_i\}_{\{1,2\}}$ is given by the two standard open affines of $\mathbb{P}^1_k$ and the boundary map is given by sending

$$d(f_0, f_1) = f_1|_{U_{ij}} - f_0|_{U_{ij}}.$$

Then the sheaf-cohomology can be calculated by the cohomology of this complex (for this purpose we might as well define it as such since we do not mention any other constructions).
The result then follows from d being surjective, since we can split a *Laurent* polynomial in $k[x, x^{-1}]$ uf into its polynomial components in the variable x and x^{-1}. Additionally we can again verify (the already known fact) $\mathrm{H}^0(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1_k}) \simeq k$ since the difference of two polynomials $f_1 - f_0$ in variables x^{-1} and x is only zero if they are constant.

Recall, that here $C(\mathbb{C})$ is the Riemann sphere, which is a topological surface of genus 0.

The arithmetic genus can be negative when for example computing the Euler characteristic with respect to a field smaller than the basefield. One can observe that $p_a(\mathbb{P}^1_{\mathbb{Q}(i)}/\mathbb{Q}) = -1$ while $p_a(\mathbb{P}^1_{\mathbb{Q}(i)}/\mathbb{Q}(i)) = 0$ as above.

2 Constructing the Blow-up

Mathematicians get arrested at some point ...
... because they blow up a plane at some point.

The idea can be directly described by giving the universal property that the Blow-up should fulfill. After that we will explicitly construct a solution to the universal property.

Let X be a scheme. Given a closed immersion $Z \to X$, the ideal sheaf $\mathcal{I}$ is not necessarily a line bundle, i.e. Z is not a Cartier divisor on X. But one might hope for the existence of a scheme (morphism) $Y \to X$ such that $Z \times_Y X \to Y$ is a Cartier divisor. Moreover the blow-up should be universal in the sense that Y should be terminal beneath such schemes (every such morphism factors through the blowup), i.e. $Bl_Z X$ is (up to unique iso) the final object in the full subcategory of $Y \to X$ in $\mathrm{Sch}_{/X}$ such that the preimage of Z is a Cartier divisor.

We shall show that such an object, satisfying the universal property actually exist by first making the following construction(s):

Definition 2.1. (*Recall*) (graded) quasi-coherent sheaf of $\mathcal{O}_X$-algebras:
A *quasi-coherent sheaf $\mathcal{O}_X$-algebras* is a quasi-coherent sheaf $\mathcal{A}$ equipped with a multiplication- and unitmorphism $\mu : \mathcal{A} \otimes_{\mathcal{O}_X} \mathcal{A} \to \mathcal{A}$ and $\eta : \mathcal{O}_X \to \mathcal{A}$ satisfying associative and (left/right) unit laws (*see* monoid object in a monoidal category).

In other words it is a quasi-coherent sheaf of *rings* that has the structure of an $\mathcal{O}_X$-algebra given by $\eta : \mathcal{O}_X \to \mathcal{A}$ making $\mathcal{A}|_{U_i}$ an A_i-algebra for every affine open $U_i = \operatorname{Spec} A_i$ of X.

Lets just call them $\mathcal{O}_X$-algebras since we are only concerned with quasi-coherent ones.

A ($\mathbb{Z}_{\geq 0}$-)*graded* sheaf of $\mathcal{O}_X$-algebras is a sheaf of $\mathcal{O}_X$-algebras $\mathcal{B}$ that admits a grading in the usual sense, i.e. $\mathcal{B} = \oplus_{n\geq 0}\mathcal{B}_n$ where each $\mathcal{B}_n$ is a quasi-coherent sub-$\mathcal{O}_X$-module (not sub-algebras since the multiplication changes degree).

Lemma/Definition 2.2. The relative (projective) spectrum:
Given a sheaf of $\mathcal{O}_X$-algebras $\mathcal{A}$, or a graded sheaf of $\mathcal{O}_X$-algebras $\mathcal{B}$ on a scheme X, there exists a unique scheme $\pi : \underline{\operatorname{Spec}}_X(\mathcal{A}) := Y \to X$, respectively $\pi : \underline{\operatorname{Proj}}_X(\mathcal{B}) := Y \to X$, such that for any affine open subscheme U of X, we have an isomorphism of U-schemes $\alpha_U : Y \times_X U \xrightarrow{\simeq} \operatorname{Spec}\mathcal{A}(U)$, respectively $\beta_U : Y \times_X U \xrightarrow{\simeq} \operatorname{Proj}\mathcal{B}(U)$ that commute with restrictions.
We call these schemes the *relative (affine) spectrum* of $\mathcal{A}$ and *relative projective spectrum* of $\mathcal{B}$.

Proof. The proof for Spec and Proj are analogue.
Given an affine cover $U_i = \operatorname{Spec} A_i$ of X we can construct $U_i' := \operatorname{Spec}\mathcal{A}(U_i) \to U_i$. The construction of these open affines will commute with restrictions, that is for $i : V \hookrightarrow W$ we have the restriction $\mathcal{A}(W) \to \mathcal{A}(V) \simeq \mathcal{A}(W) \otimes_{\mathcal{O}_{U_i}(W)} \mathcal{O}_{U_i}(V)$ (by q-coh) inducing a pullback diagram along i.
In particular this is true for the inclusions $U_i \cap U_j \hookrightarrow U_i$ and $U_i \cap U_j \hookrightarrow U_j$ which gives an isomorphism $U_i'|_{U_i\cap U_j} \xrightarrow{\simeq} U_j'|_{U_i\cap U_j}$ since the tensor products are isomorphic. One can also verify the cocycle condition and hence we can glue the schemes U_i' to the desired scheme. □

Remark. In the affine case, where an $\mathcal{O}_X$-algebra is equivalent to an A-algebra this recovers the usual Spec and Proj hence we might just write $\operatorname{Spec}\mathcal{A}$ and $\operatorname{Proj}\mathcal{B}$ when the base is clear from the context. Moreover $\pi_*\mathcal{O}_{\underline{\operatorname{Spec}}_X(\mathcal{A})} \simeq \mathcal{A}$ generalizing the adjunction $\Gamma \dashv \operatorname{Spec}$ to schemes affine over X.

Example. (not needed for this talk):
We can make sense of the name "Vector-bundle" for finite locally free sheaves in a more "topological" way (remember that in topology a vector bundle is given by a second topological space that projects down to the base space $E \to X$ whereas in algebraic geometry a vector bundle is a sheaf on the base space X). One might wonder if this sheaf is also represented by an actual scheme $E \to X$.
In the case of the trivial line bundle $\mathcal{O}_X$ this might be obvious, since this functor is represented by $\mathbb{A}^1_X$ (sending x to the section). To generalize this for general locally free sheaves by finding a way to associate such a space we can make the following construction called the *total-space* of $\mathcal{E}$:

$$\mathbb{V}(\mathcal{E}) := \underline{\operatorname{Spec}}_X(\operatorname{Sym}_{\mathcal{O}_X}\mathcal{E}^\vee)$$

which recovers affine locally the affine-n-space over a trivializing cover $U = \operatorname{Spec} A$ of $\mathcal{E}$ by:

$$\operatorname{Spec}\operatorname{Sym}((A^{\oplus n})^\vee) \cong \operatorname{Spec}\operatorname{Sym}(A^\vee)^{\oplus n} \cong \operatorname{Spec} A[x_1, ..., x_n] =: \mathbb{A}^n_U.$$

Recall that $\operatorname{Sym} V := (\oplus_{k=0}^\infty V^{\otimes k})/(v \otimes w - w \otimes v)$. It can be viewed as free construction i.e. a left adjoint to the forgetful functor $U : \mathsf{CAlg}_R \to \mathsf{Mod}_R$ from which we obtain the above isomorphism.
Example.
Given a vector bundle (or coherent sheaf) $\mathcal{M}$ on X, since $\operatorname{Sym}_{\mathcal{O}_X}\mathcal{M}$ is graded we can also from:

$$\mathbb{P}(\mathcal{M}) := \underline{\operatorname{Proj}}_X(\operatorname{Sym}_{\mathcal{O}_X}\mathcal{M}) \text{ (or } \underline{\operatorname{Proj}}_X(\operatorname{Sym}_{\mathcal{O}_X}\mathcal{M}^\vee) \text{ depending on the convention)}$$

which parameterizes rank-1 quotients of $\mathcal{M}$ (or rank-1 subbundles of $\mathcal{M}$) and hence recovers $\mathbb{P}^n_X$ when $\mathcal{M} = \mathcal{O}_X^{\oplus n}$.
Remark. $\mathcal{O}_Y(n)$
On the projective space $\mathbb{P}(\mathcal{O}_X^{\oplus n})$ we know the line bundle $\mathcal{O}(1)$. We can generalize this to general relative Proj-constructions where the $\mathcal{O}_X$-algebra is not given by the above symmetric algebra.
This is done by gluing the sheaves $\mathcal{O}(1)$ for each $\operatorname{Proj}\mathcal{B}(U_i)$ over an open cover $U_i \subset X$.
We call the resulting sheaf $\mathcal{O}_Y(1)$ for $Y = \operatorname{Proj}\mathcal{B}$ and define the *twists* of the structure sheaf $\mathcal{O}_Y(n) := \mathcal{O}_Y(1)^{\otimes n}$.

Definition 2.3. The Blow-up scheme:
Given a scheme X together with a closed immersion $Z \to X$ with ideal sheaf $\mathcal{I}$, we define the *Blow-up of X along Z* by

$$\tilde{X} := Bl_Z X := \underline{\mathrm{Proj}}_X(\bigoplus_{d\geq 0} \mathcal{I}^d) \to X$$

where the $\mathcal{O}_X$-algebra is by construction generated by the degree 1 component $\mathcal{I}$ over the degree 0 componen $\mathcal{I}^0 \simeq \mathcal{O}_X$ also called the *Rees algebra* of $\mathcal{I}$.
Notice that we can view $\oplus_{d\geq 0} I^d \simeq A \oplus t \cdot I \oplus t^2 \cdot I^2 \oplus ... =: A[It] \subseteq A[t]$.

Since Proj can be covered via $D_+(a)$'s for degree-1 generators, we can also cover $\operatorname{Proj} \oplus_{d\geq 0} I^d$ by the spectra of the so called *affine blowup algebras* $A[\frac{I}{a}] := (\oplus_{d\geq 0} I^d)_{(a)}$ for degree-1 $a \in I \subset \oplus_{d\geq 0} I^d$.

Lemma/Definition 2.4. The exeptional divisor:
Moreover we have an isomorphism between the *exeptional divisor* E of $Bl_Z X$ (we will see, that it indeed forms a Cartier divisor) and the following relative-proj:

$$E \simeq \underline{\mathrm{Proj}}_Z(\bigoplus_{d\geq 0} \mathcal{I}^d/\mathcal{I}^{d+1})$$

where $E := b^{-1}(Z) := Bl_Z X \times_X Z$.

Proof. Affine locally we know that $Bl_Z X \times_X Z \simeq \operatorname{Proj} \oplus_{d\geq 0}(I^d \otimes_A A/I)$ (tensoring commutes with direct sums). But $I^d \otimes_A A/I \simeq I^d/I \cdot I^d = I^d/I^{d+1}$. Hence E also admits a closed immersion into $Bl_Z X$. □

Lemma 2.5. *The fiber over the compliment $b|_{b^{-1}(X-Z)} : b^{-1}(X-Z) \to X - Z$ is an isomorphism. Moreover the exeptional divisor E of $b : \tilde{X} := Bl_Z X \to X$ is an effective Cartier divisor on $\tilde{X}$ with invertible sheaf of ideals $\mathcal{O}_{\tilde{X}}(-E) = b^{-1}\mathcal{I} \cdot \mathcal{O}_{\tilde{X}} \simeq \mathcal{O}_{\tilde{X}}(1)$ or the dual $\mathcal{O}_{\tilde{X}}(E) \simeq \mathcal{O}_{\tilde{X}}(-1)$.*

Proof. Since the relative Proj is constructed locally on the base, we can assume that we blow up of X along $\varnothing$ where $\mathcal{I} = \mathcal{O}_X$, i.e. blowing up $X - Z$.
In this case $\oplus_{d\leq 0} \mathcal{O}_X(X)^d = \mathcal{O}_X(X) \oplus x \cdot \mathcal{O}_X(X) \oplus x^2 \cdot \mathcal{O}_X(X) \oplus ... = \mathcal{O}_X(X)[x]$ as a graded algebra. But $\operatorname{Proj} A[x] \simeq \operatorname{Spec} A$ for $A[x]$ generated by x (in degree 1) as an A-algebra.

To show that E is an effective cartier divisor, we again use the fact, that $Bl_Z X$ is constructed by gluing $\operatorname{Proj} \oplus_{d\geq 0} I_i^d$ over an open affine cover $\operatorname{Spec} A_i$ of X. We can further refine that cover by an open affine cover for each such projective space given by the spectra of the affine Blowup-algebras $U_{i,a} = \operatorname{Spec} A_i[\frac{I_i}{a}]$ (lets fix i and just write I and A).
Now $E|_{U_{i,a}} = \operatorname{Spec} A[\frac{I}{a}]/IA[\frac{I}{a}]$. Hence we need to show that $IA[\frac{I}{a}]$ is principally generated. Indeed we can verify $IA[\frac{I}{a}] = aA[\frac{I}{a}]$. The inclusion $aA[\frac{I}{a}] \subseteq IA[\frac{I}{a}]$ is immediate from $a \in I$. The other inclusion $IA[\frac{I}{a}] \subseteq aA[\frac{I}{a}]$ we observe that elements in $A[\frac{I}{a}]$ can be represented by x/a^n with $x \in I^n$ with the usual relation of the localization so a generator of $IA[\frac{I}{a}]$ is of the form x/a^n with $x \in I^{n+1}$. Now $x/a^n = a(x/a^{n+1})$ where $x/a^{n+1} \in A[\frac{I}{a}]$.

For the line bundle, observe that the degree 1 part of the localization $((\oplus_{d\geq 0} I^d)_a)_1$ also consists of elements x/a^k where $x \in I^{k+1}$. □

Lemma 2.6. *The Blow-up $Bl_Z X =: \tilde{X}$ satisfies the above described universal property. That means for any morphism $f : W \to X$ such that $f^{-1}Z$ is an effective cartier divisor there exists a unique morphism $g : W \to \tilde{X}$ making the following diagram commutative:*

$$\begin{array}{ccc} W & \xrightarrow{\exists! g} & \tilde{X} \\ & \searrow^{f} & \downarrow b \\ & & X \end{array}$$

Proof. For this fix such a $f : W \to X$ and call $D := f^{-1}Z$. Furthermore let $\mathcal{I}_Z$ denote the ideal sheaf corresponding to Z and $\mathcal{I}_D$ of D which is invertible by assumption.

We just need to construct any such morphism $g : W \to \tilde{X}$, since uniqueness will then follow from the following facts:

The restriction $g|_{W-D} : W - D \to \tilde{X} - E$ is uniquely determined since it will equal the restriction $f|_{W-D} : W - D \to X - Z$ (by the first part of 2.5 $X - Z \cong b^{-1}(X - Z) \cong \tilde{X} - E$)

The other fact is, that $W - D$ is scheme-theoretically dense in W (in the affine case this is because the effective Cartier divisor is given by a non-zerodivisor $f \in A$ hence the closure $\operatorname{Spec}(A/\ker(i : A \hookrightarrow A_f))$ is just $\operatorname{Spec} A$). Since the projective space is seperated we get by (*cf.*: [Sta23], Tag 01RH, Lemma 29.7.10.; the equalizer of two maps is a closed subscheme), that the restricted map on this open sense subscheme determines the original map uniquely.

We shall construct the desired $g : W \to \tilde{X}$ by first assuming that the target is the blow up of an affine base $X = \operatorname{Spec} B$. Let furthermore $U_i = \operatorname{Spec} A_i$ be an affine cover of W.
In this situation we have ring morphisms $B \to \oplus_{d\geq 0} I^d$ and $f_i^* : B \to A_i$. By assumption we also know that $\mathcal{I}_D$ is invertible, so $I_D A_i = a_i A_i$ with $a_i \in A_i$ a non-zerodivizor so $f_i^*(b \in I^d) \in (a_i^d)A_i$. We can then construct a morphism of graded B-algebras $g_i^* : R := \oplus_{d\geq 0} I^d \to A_i[x]$ by sending degree d elements $I^d \ni b \mapsto x^d(f_i^*(b)/a_i^d)$ with $(g_i^*(R_+ := \oplus_{d>0} I^d)) = (x) =: A_i[x]_+$.

We also know that $\operatorname{Spec} A_i \simeq \operatorname{Proj} A_i[x]$, and hence this will give us the desired morphism of $\operatorname{Spec} B$-schemes $\operatorname{Spec} A_i \to \operatorname{Proj} \oplus_{d\geq 0} I^d$.

□

Lemma 2.7. *When the defining sheaf of ideals $\mathcal{I}$ is of finite type, the Blow-up $b : \tilde{X} \to X$ is a projective morphism in the general sense of factoring through a closed immersion in some $\mathbb{P}(\mathcal{E}) \to X$ where $\mathcal{E}$ is a quasi-coherent finite type $\mathcal{O}_X$-module.*

Proof. Recall $\mathbb{P}(\mathcal{E}) := \underline{\operatorname{Proj}}_X(\operatorname{Sym}_{\mathcal{O}_X} \mathcal{E})$. Hence, to get a morphism over X we just need to construct a surjective morphism of $\mathcal{O}_X$-algebras $\operatorname{Sym}_{\mathcal{O}_X} \mathcal{E} \to \oplus_{d\geq 0} \mathcal{I}^d$. Indeed the canonical morphism $\operatorname{Sym}_{\mathcal{O}_X} \mathcal{I} \to \oplus_{d\geq 0} \mathcal{I}^d$; $v_1 \otimes v_2 \otimes ... \otimes v_d \mapsto v_1 v_2 ... v_d$ is already surjective hence we get the closed immersion $\tilde{X} \to \operatorname{Proj}_X(\operatorname{Sym}_{\mathcal{O}_X} \mathcal{I}) =: \mathbb{P}(\mathcal{I})$. □

Remark. Similarly, we could show that E is projective over Z:

$$\begin{array}{ccccc} Bl_Z X = \operatorname{Proj}_X \oplus_{d\geq 0} \mathcal{I}^d & \xhookrightarrow{\iota_1} & \mathbb{P}(\mathcal{I}) & \longrightarrow & X \\ \uparrow & & \uparrow & & \uparrow \\ E \simeq \operatorname{Proj}_Z \oplus_{d\geq 0} \mathcal{I}^d/\mathcal{I}^{d+1} & \xhookrightarrow{\iota_2} & \mathbb{P}(\mathcal{I}/\mathcal{I}^2) & \longrightarrow & Z \end{array}$$

If $\mathcal{I}$ is locally generated by a regular sequence, we also have the immersions ι_1 and ι_2 are both isomorphisms. Hence under this assumption the exeptional divisor E can be identified with the projectivization of the so called *conormal sheaf* $\mathcal{C}_{Z/X} := \mathcal{I}/\mathcal{I}^2$ parameterizing "first-order directions transverse to Z in X".

References

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3 Exercises

1. Calculate the Blow-up of the "cross" $X = \operatorname{Spec} k[x,y]/(xy)$ at the origin $\mathfrak{I} = (x,y)$.
2. Calculate the Blow-up of the plane $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$ at the origin $\mathfrak{I} = (x,y)$.
3. Calculate the Blow-up of $\mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x]$ at $\mathfrak{I} = (p,x)$.

Since the solution paths will be discussed in presence, here we will only give the solutions:

1. $Bl_Z(X) \cong \operatorname{Proj} k[x,y,u,v]/(xy, xv - yu, uv) \cong \mathbb{A}^1_k \coprod \mathbb{A}^1_k$
   and $E \cong \operatorname{Proj} k[u,v]/(uv) \cong \operatorname{Spec} k \coprod \operatorname{Spec} k$.
2. $Bl_Z(X) \cong \operatorname{Proj} k[x,y,u,v]/(xv - yu)$ and $E \cong \mathbb{P}^1_k$.
3. $Bl_Z(X) \cong \operatorname{Proj} \mathbb{Z}[x,u,v]/(pv - xu)$ and $E \cong \mathbb{P}^1_{\mathbb{F}_p}$.